Session-7-exercises.R

```
#################
### Session 7 ###
#################

library(tidyverse)
library(lubridate)

# The time started at
origin

## [1] "1970-01-01 UTC"

# The date today
today() # equivalent to Sys.Date() in base

## [1] "2026-01-04"

as.numeric(today())

## [1] 20457

# The time now
now() # equivalent to Sys.time() in base

## [1] "2026-01-04 01:20:19 CET"

as.numeric(now())

## [1] 1767486019

# We can also use lubridate to create date values out of different strings
mdy("01/12/2026")

## [1] "2026-01-12"

mdy("01:12:26")

## [1] "2026-01-12"

dmy("12/01/26")

## [1] "2026-01-12"

ymd("2026-01-12")

## [1] "2026-01-12"

# And we can include times
ymd_hms("2026-01-12 10:30:26")

## [1] "2026-01-12 10:30:26 UTC"
```

```r
# Let's set that to Eastern time
ymd_hms("2026-01-12 10:30:26", tz='EST')

## [1] "2026-01-12 10:30:26 EST"

# We can  extract some derived values such as the weekday
wday("2026-01-12")

## [1] 2

# or day of the year
yday("2026-01-12")

## [1] 12

# Play with alan
alan <- mdy_hm("June 23, 1912, 02:15 AM", tz='Europe/London')

wday(alan)

## [1] 1

wday(alan, label=TRUE)

## [1] Sun
## Levels: Sun < Mon < Tue < Wed < Thu < Fri < Sat

wday(alan, label = TRUE, abbr = FALSE) # Lubridate is a bit more complex

## [1] Sunday
## 7 Levels: Sunday < Monday < Tuesday < Wednesday < Thursday < ... <
Saturday

# Lubridate provides semester, am/pm, daylight savingtime, leap_year
semester(alan)

## [1] 1

am(alan)

## [1] TRUE

dst(alan)

## [1] FALSE

leap_year(alan)

## [1] TRUE

# add
year(alan) + 3

## [1] 1915
```

```r
tz(alan)

## [1] "Europe/London"

#What was the time in Tokyo?
grep("Tokyo", OlsonNames(), value = TRUE)

## [1] "Asia/Tokyo"

with_tz(alan, "Asia/Tokyo") # Japan standard time

## [1] "1912-06-23 11:15:00 JST"

tz(alan)

## [1] "Europe/London"

# Zeit, bitte bleib stehen, bleib stehen
floor_date(now(), unit = "month") # round down

## [1] "2026-01-01 CET"

floor_date(now(), unit = "year")

## [1] "2026-01-01 CET"

round_date(now(), unit = "hour") # round to nearest unit

## [1] "2026-01-04 01:00:00 CET"

ceiling_date(now(), unit = "minutes") # round up

## [1] "2026-01-04 01:21:00 CET"

#Time differences
ddays(370)

## [1] "31968000s (~1.01 years)"

days(370)

## [1] "370d 0H 0M 0S"

# how to calculate with Lubridate
LunarOrbit <- ddays(29) + dhours(12) + dminutes(44)

# calculate a date + lunar orbit
ymd("2024-01-31") + LunarOrbit

## [1] "2024-02-29 12:44:00 UTC"

ymd("2025-01-31") + LunarOrbit

## [1] "2025-03-01 12:44:00 UTC"
```

```r
dmy("28 February, 2024") + years(1)

## [1] "2025-02-28"

dmy("29 February, 2024") + years(1)

## [1] NA

dmy("29 February, 2024") + dyears(1) # One "dyear" is 365.25 days

## [1] "2025-02-28 06:00:00 UTC"

dmy("29 February, 2024") + months(1)

## [1] "2024-03-29"

dmy("31 January, 2024") + months(1)

## [1] NA

dmy("31 January, 2024") + dmonths(1) # One "dmonth" is 1/12 "dyear"

## [1] "2024-03-01 10:30:00 UTC"

# Lubridate intervals - a fixed start time and a fixed end time

alandiff <- interval(alan, now())
alandiff/years(1) # periods understand leap years

## [1] 113.5341

alandiff/dyears(1) # durations don't

## [1] 113.533

round(alandiff/dyears(1))

## [1] 114

##############################################################"
```

```
#Exercise 7.1
library(tidyverse)
library(lubridate)

# Read the dataset
weather <- read_csv("mexicanweather.csv")

weather <- weather %>%
  pivot_wider(names_from = element, values_from=value)

# Let's look at what we have
weather
```

```
## # A tibble: 16,871 × 4
##    station     date        TMAX  TMIN
##    <chr>       <date>     <dbl> <dbl>
##  1 MX000017004 1955-04-01   310   150
##  2 MX000017004 1955-05-01   310   200
##  3 MX000017004 1955-06-01   300   160
##  4 MX000017004 1955-07-01   270   150
##  5 MX000017004 1955-08-01   230   140
##  6 MX000017004 1955-09-01   230   150
##  7 MX000017004 1955-10-01   240   150
##  8 MX000017004 1955-11-01   270   130
##  9 MX000017004 1955-12-01   240   130
## 10 MX000017004 1956-01-01   240    90
## # ℹ 16,861 more rows
```

```
# We can use lubridate functions to extract elements of the date
weather <- weather %>%
  mutate(year=year(date), month=month(date), day=day(date))
weather
```

```
## # A tibble: 16,871 × 7
##    station     date        TMAX  TMIN  year month   day
##    <chr>       <date>     <dbl> <dbl> <dbl> <dbl> <int>
##  1 MX000017004 1955-04-01   310   150  1955     4     1
##  2 MX000017004 1955-05-01   310   200  1955     5     1
##  3 MX000017004 1955-06-01   300   160  1955     6     1
##  4 MX000017004 1955-07-01   270   150  1955     7     1
##  5 MX000017004 1955-08-01   230   140  1955     8     1
##  6 MX000017004 1955-09-01   230   150  1955     9     1
##  7 MX000017004 1955-10-01   240   150  1955    10     1
##  8 MX000017004 1955-11-01   270   130  1955    11     1
##  9 MX000017004 1955-12-01   240   130  1955    12     1
## 10 MX000017004 1956-01-01   240    90  1956     1     1
## # ℹ 16,861 more rows
```

```
unique(weather$station)
```

```
## [1] "MX000017004"
```

```
# There is only one station, it is not a variable,
# we can remove from the database.
```

```r
# Exercise 7.2

#a) Store your birth time in an object mybirth. Pay attention to the right
time zone.
mybirth <- mdy_hm("June 28, 1971, 07:30 AM", tz='Africa/Johannesburg')

#b) Check if you were born in a leap year, or daylight-saving time.
leap_year(mybirth)

## [1] FALSE

dst(mybirth)

## [1] FALSE

#c) What day of the week?
wday(mybirth, label = TRUE, abbr= FALSE)

## [1] Monday
## 7 Levels: Sunday < Monday < Tuesday < Wednesday < Thursday < ... <
Saturday

#d) Calculate your age in duration and period. Explain the difference between
them.
mydiff <- interval(mybirth, now())
mydiff/years(1) # periods understand leap years

## [1] 54.51996

mydiff/dyears(1) # durations don't

## [1] 54.52097

#e) Calculate when you will be 20,000 days old.
mybirth + days(20000)

## [1] "2026-03-31 07:30:00 SAST"

#f) What will be the time in Auckland then?
grep("Auckland", OlsonNames(), value = TRUE)

## [1] "Pacific/Auckland"

with_tz(mybirth + days(20000), tz="Pacific/Auckland")

## [1] "2026-03-31 18:30:00 NZDT"
```

```r
# Exercise 7.3
library(tidyverse)
#a)
survey <- read_tsv("Survey Data For My Best Exam 2024.txt")

#b)
# do NOT
# library(janitor)
# survey <- survey %>% remove_empty(which = c("cols"))
# otherwise, it can be a great solution

summary(survey)
```

```
##   Timestamp         How old are you?     Industry          Job title
##  Length:27596       Length:27596       Length:27596       Length:27596
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##  Additional context on job title Annual salary      Other monetary comp
##  Length:27596                    Length:27596       Length:27596
##  Class :character                Class :character   Class :character
##  Mode  :character                Mode  :character   Mode  :character
##    Currency         Currency - other   Additional context on income
##  Length:27596       Length:27596       Length:27596
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##    Country              State               City
##  Length:27596       Length:27596       Length:27596
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##  Years of experience in field Overall years of professional experience
##  Length:27596                 Length:27596
##  Class :character             Class :character
##  Mode  :character             Mode  :character
##  Highest level of education completed    Gender              Race
##  Length:27596                         Length:27596       Length:27596
##  Class :character                     Class :character   Class :character
##  Mode  :character                     Mode  :character   Mode  :character
##   ...19          ...20          ...21          ...22          ...23
##  Mode:logical   Mode:logical   Mode:logical   Mode:logical   Mode:logical
##  NA's:27596     NA's:27596     NA's:27596     NA's:27596     NA's:27596
##
##   ...24
##  Mode:logical
##  NA's:27596
##
```

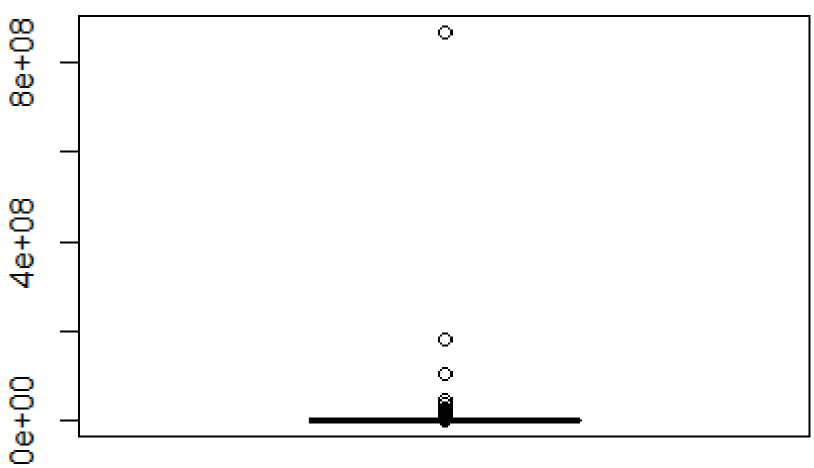
```r
# Columns 19:24 have as many NAs as data entries other columns
survey <- survey[,1:18]
summary(survey)
```

```
##   Timestamp          How old are you?     Industry           Job title
##  Length:27596       Length:27596       Length:27596       Length:27596
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##  Additional context on job title Annual salary      Other monetary comp
##  Length:27596                    Length:27596       Length:27596
##  Class :character                Class :character   Class :character
##  Mode  :character                Mode  :character   Mode  :character
##    Currency          Currency - other   Additional context on income
##  Length:27596       Length:27596       Length:27596
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##    Country             State               City
##  Length:27596       Length:27596       Length:27596
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##  Years of experience in field Overall years of professional experience
##  Length:27596                 Length:27596
##  Class :character             Class :character
##  Mode  :character             Mode  :character
##  Highest level of education completed    Gender              Race
##  Length:27596                         Length:27596       Length:27596
##  Class :character                     Class :character   Class :character
##  Mode  :character                     Mode  :character   Mode  :character
```

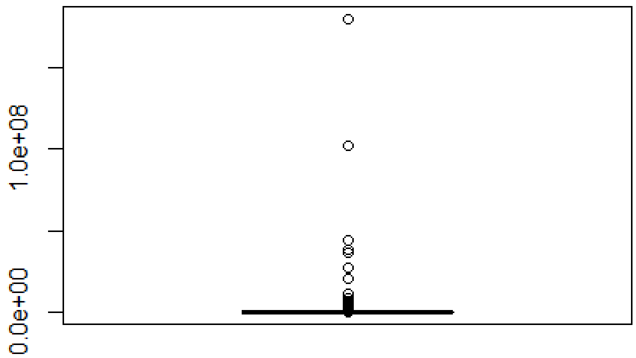
```r
glimpse(survey)
```

```
## Rows: 27,596
## Columns: 18
## $ Timestamp                                  <chr> "4/27/2021 11:02:10", "4/27…
## $ `How old are you?`                         <chr> "25-34", "25-34", "25-34", …
## $ Industry                                   <chr> "Education (Higher Educatio…
## $ `Job title`                                <chr> "Research and Instruction L…
## $ `Additional context on job title`          <chr> NA, NA, NA, NA, NA, NA, NA,…
## $ `Annual salary`                            <chr> "55 000", "54 600", "34 000…
## $ `Other monetary comp`                      <chr> "0", "4000", NA, "3000", "7…
## $ Currency                                   <chr> "USD", "GBP", "USD", "USD",…
## $ `Currency - other`                         <chr> NA, NA, NA, NA, NA, NA, NA,…
## $ `Additional context on income`             <chr> NA, NA, NA, NA, NA, NA, NA,…
## $ Country                                    <chr> "United States", "United Ki…
## $ State                                      <chr> "Massachusetts", NA, "Tenne…
## $ City                                       <chr> "Boston", "Cambridge", "Cha…
## $ `Years of experience in field`             <chr> "5-7 years", "8 - 10 years"…
## $ `Overall years of professional experience` <chr> "5-7 years", "5-7 years", "…
## $ `Highest level of education completed`     <chr> "Master's degree", "College…
## $ Gender                                     <chr> "Woman", "Non-binary", "Wom…
## $ Race                                       <chr> "White", "White", "White", …
```

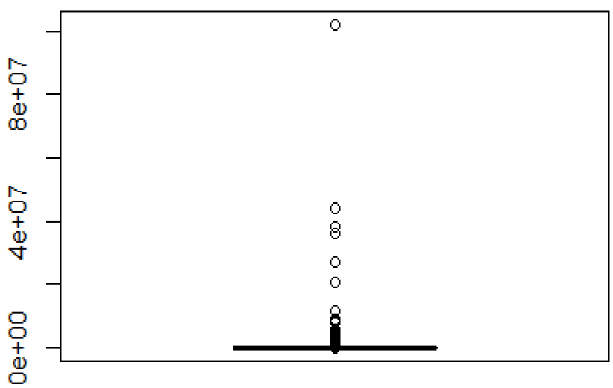
```r
#c)
survey <- survey %>% rename(time = Timestamp) %>%
  mutate(time = mdy_hms(time))

sum(is.na(survey$time))

## [1] 0

#d)
OlsonNames()

EShour <- hour(with_tz(now(), tzone = "America/El_Salvador"))

#or
EShour <- hour(with_tz(now(),
                       tzone = grep("Salvador", OlsonNames(), value = TRUE,
                                    ignore.case = TRUE)))

#e)
survey$`Annual salary` <- str_replace_all(survey$`Annual salary`, " ", "")
survey <- survey %>%
  mutate(salary = as.numeric(`Annual salary`))
sum(is.na(survey$salary))

## [1] 0

boxplot(survey$salary)
```



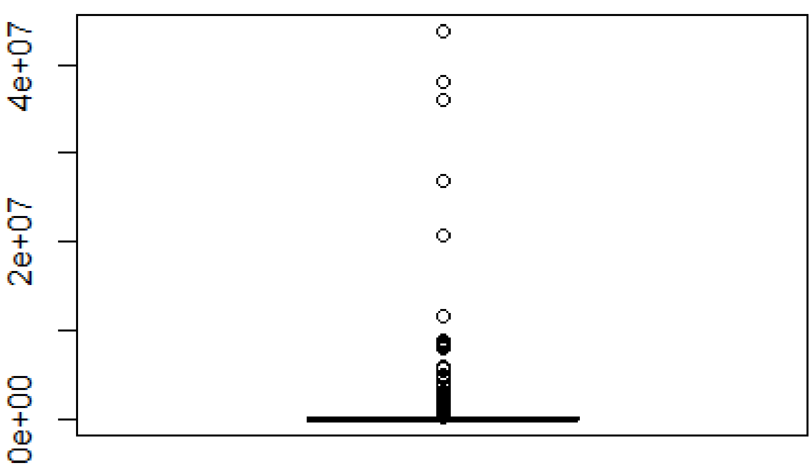
```r
survey <- survey[survey$salary!=max(survey$salary),]
boxplot(survey$salary)
```




```r
survey <- survey[survey$salary!=max(survey$salary),]
boxplot(survey$salary)
```




```r
survey <- survey[survey$salary!=max(survey$salary),]
boxplot(survey$salary)
```




```r
#f)
glimpse(survey)
```

```
## Rows: 27,593
## Columns: 19
## $ time                                        <dttm> 2021-04-27 11:02:10, 2021-…
## $ `How old are you?`                          <chr> "25-34", "25-34", "25-34", …
## $ Industry                                    <chr> "Education (Higher Educatio…
## $ `Job title`                                 <chr> "Research and Instruction L…
## $ `Additional context on job title`           <chr> NA, NA, NA, NA, NA, NA, NA,…
## $ `Annual salary`                             <chr> "55000", "54600", "34000", …
## $ `Other monetary comp`                       <chr> "0", "4000", NA, "3000", "7…
## $ Currency                                    <chr> "USD", "GBP", "USD", "USD",…
## $ `Currency - other`                          <chr> NA, NA, NA, NA, NA, NA, NA,…
## $ `Additional context on income`              <chr> NA, NA, NA, NA, NA, NA, NA,…
## $ Country                                     <chr> "United States", "United Ki…
## $ State                                       <chr> "Massachusetts", NA, "Tenne…
## $ City                                        <chr> "Boston", "Cambridge", "Cha…
## $ `Years of experience in field`              <chr> "5-7 years", "8 - 10 years"…
## $ `Overall years of professional experience` <chr> "5-7 years", "5-7 years", "…
## $ `Highest level of education completed`      <chr> "Master's degree", "College…
## $ Gender                                      <chr> "Woman", "Non-binary", "Wom…
## $ Race                                        <chr> "White", "White", "White", …
## $ salary                                      <dbl> 55000, 54600, 34000, 62000,…
```

```
table(survey$`Years of experience in field`, survey$`Overall years of
professional experience`)
```

```
##                    1 year or less 11 - 20 years 2 - 4 years 21 - 30 years
##   1 year or less              453             3          23             0
##   11 - 20 years               160          4963         811            14
##   2 - 4 years                 418             7        2440             0
##   21 - 30 years                26          1264         178          1545
##   31 - 40 years                 5           166          25           266
##   41 years or more              2            18           1            26
##   5-7 years                   212            18        1695             3
##   8 - 10 years                132            31         947             0
##
##                    31 - 40 years 31 - 400 years 41 years or more 5-7 years
##   1 year or less               1              0                0         7
##   11 - 20 years                1              0                3      1454
##   2 - 4 years                  3              0                2        50
##   21 - 30 years               13              0                1       258
##   31 - 40 years              314              1                2        38
##   41 years or more            38              0               29         2
##   5-7 years                    3              0                0      2813
##   8 - 10 years                 2              0                1      1799
##
##                    8 - 10 years
##   1 year or less              3
##   11 - 20 years            2112
##   2 - 4 years                15
##   21 - 30 years             306
##   31 - 40 years              39
##   41 years or more            3
##   5-7 years                  43
##   8 - 10 years             2385
```

```r
# There is a value in the 'Overall years' 31-400 years, it should be 31-40 years.
# Many people have more experience in the field than the overall, the contrary is rare.
# Almost sure that the two variables are cross-changed.
# After renaming them, we should remove people / experience values where more field experience
# than overall experience.

#g)
survey <- survey %>%
  mutate(field=`Overall years of professional experience`) %>%
  mutate(overall=`Years of experience in field`) %>%
  mutate(field=ifelse(field=="31 - 400 years", "31 - 40 years", field))

table(survey$field, survey$overall)
```

```
##                     
##                      1 year or less 11 - 20 years 2 - 4 years 21 - 30 years
##   1 year or less                453           160         418            26
##   11 - 20 years                   3          4963           7          1264
##   2 - 4 years                    23           811        2440           178
##   21 - 30 years                   0            14           0          1545
##   31 - 40 years                   1             1           3            13
##   41 years or more                0             3           2             1
##   5-7 years                       7          1454          50           258
##   8 - 10 years                    3          2112          15           306
##                     
##                      31 - 40 years 41 years or more 5-7 years 8 - 10 years
##   1 year or less                 5                2       212          132
##   11 - 20 years                166               18        18           31
##   2 - 4 years                   25                1      1695          947
##   21 - 30 years                266               26         3            0
##   31 - 40 years                315               38         3            2
##   41 years or more               2               29         0            1
##   5-7 years                     38                2      2813         1799
##   8 - 10 years                  39                3        43         2385
```